Rigshospitalet

Case presentation

From Analytics to Action

REGION H

Maia Norstved Klett

Data Analyst

Maia.norstved.klett@regionh.dk

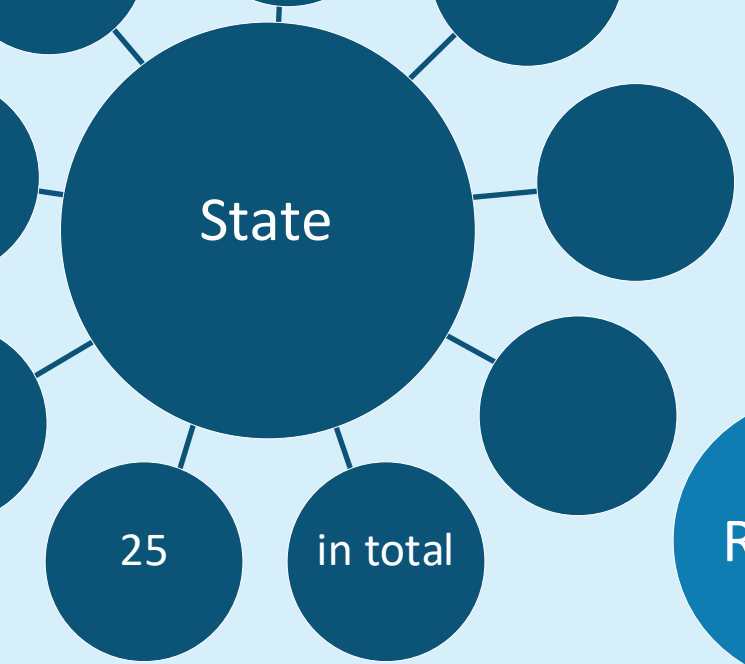
Majken Maul Leegaard

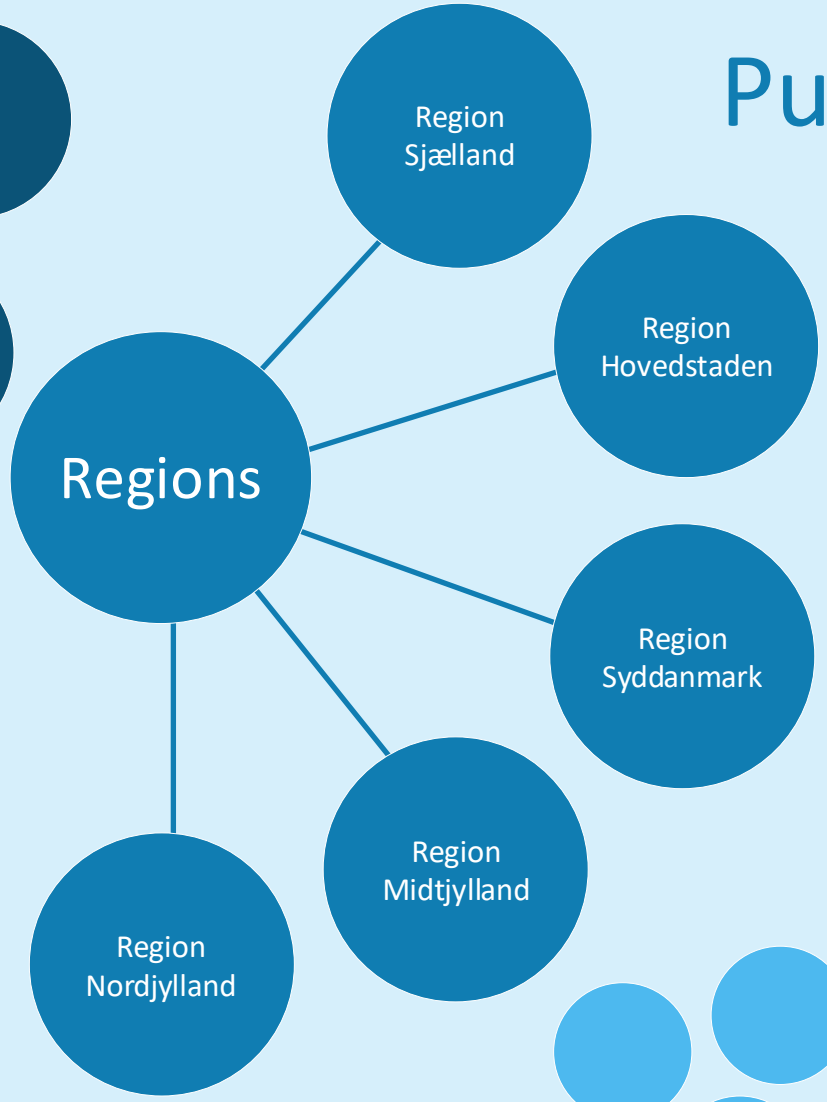
Head of Data

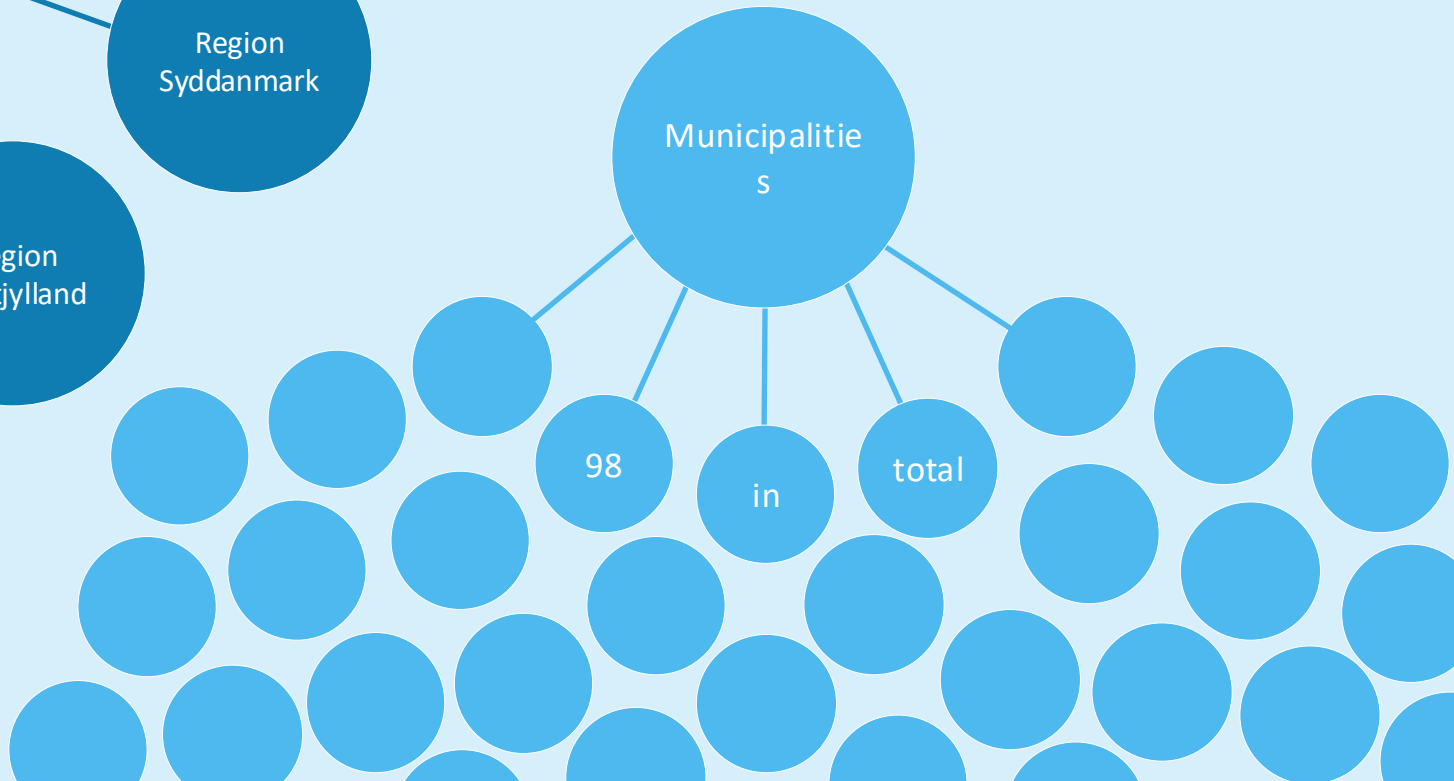
Jeanette Therming Jørgensen

Data Analyst

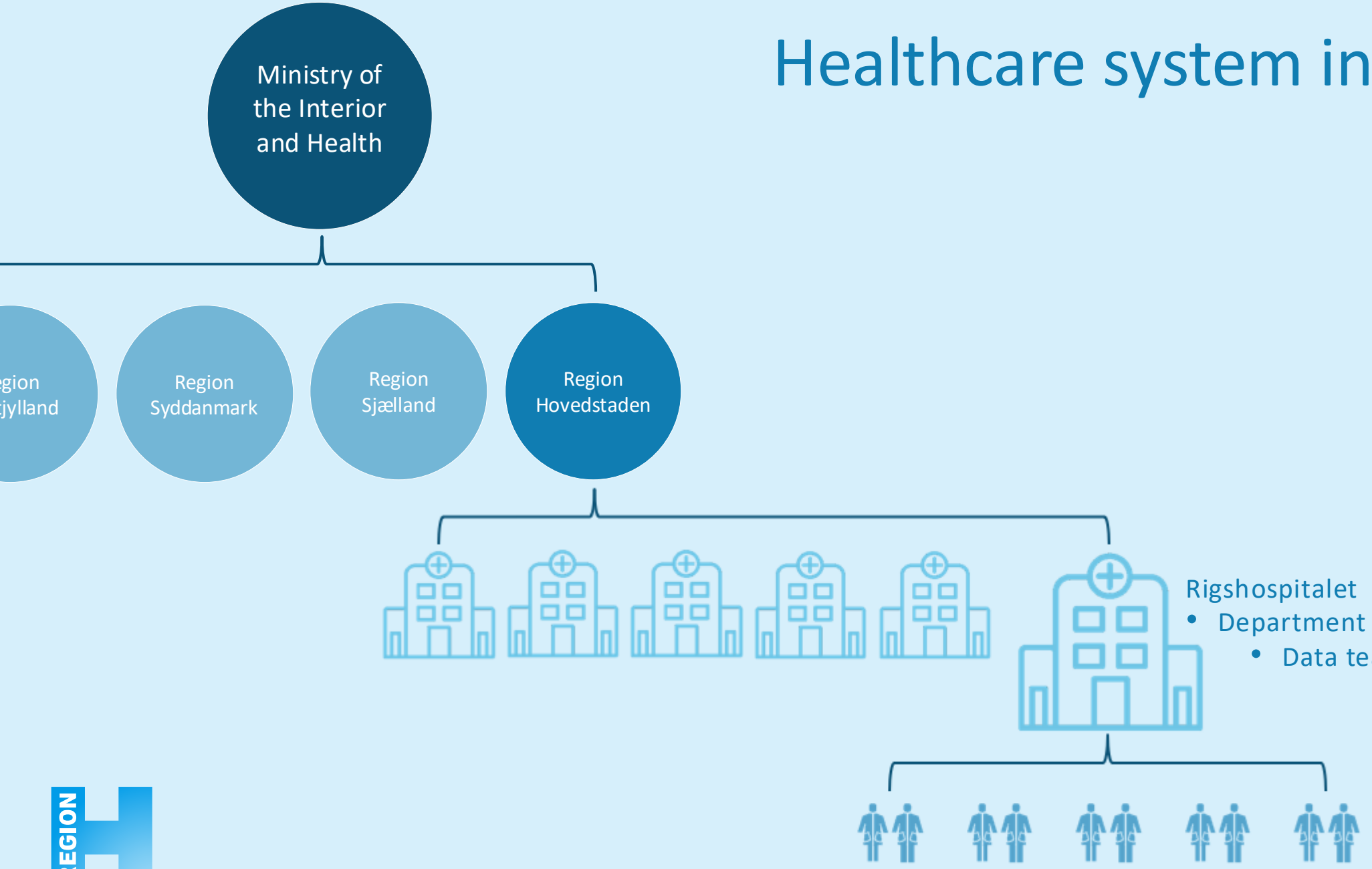
Public sector in Denmark







Healthcare system in Denmark



Rigshospitalet
- Department of Economic & Planning
  - Data team

Mission

Rigshospitalet is a highly specialized hospital for all of Denmark, with treatment, research, innovation and training at international level.

Working with international healthcare stakeholders, Rigshospitalet contributes to developing the overall healthcare system so that all patients, irrespective of the rarity or complexity of their illness, receive high-quality treatment.

Vision

To benefit the individual patient
and the overall healthcare system

Data Team

Data input

Expert & guide in registration practices

Sundhedsplatformen, LPR3 & helping ease processes

Patient billing & treatment

Organization structure

Data output

Management information & Business Information Platform

Data modelling and analysis based on returning needs and case projects

Data strategy

Sparring & collaboration

Data-driven management

1. Correct data input

- Through good registration practices

2. Management information

- Report development
- Meaningful visualizations

3. Analysis & insights

- Deep dive in data
- Identify challenges
- Focus and delimitation

4. Deliberations & decisions

- Curiosity and courage to act

5. Carry out initiatives

- Act on agreed changes
- Continuous focus

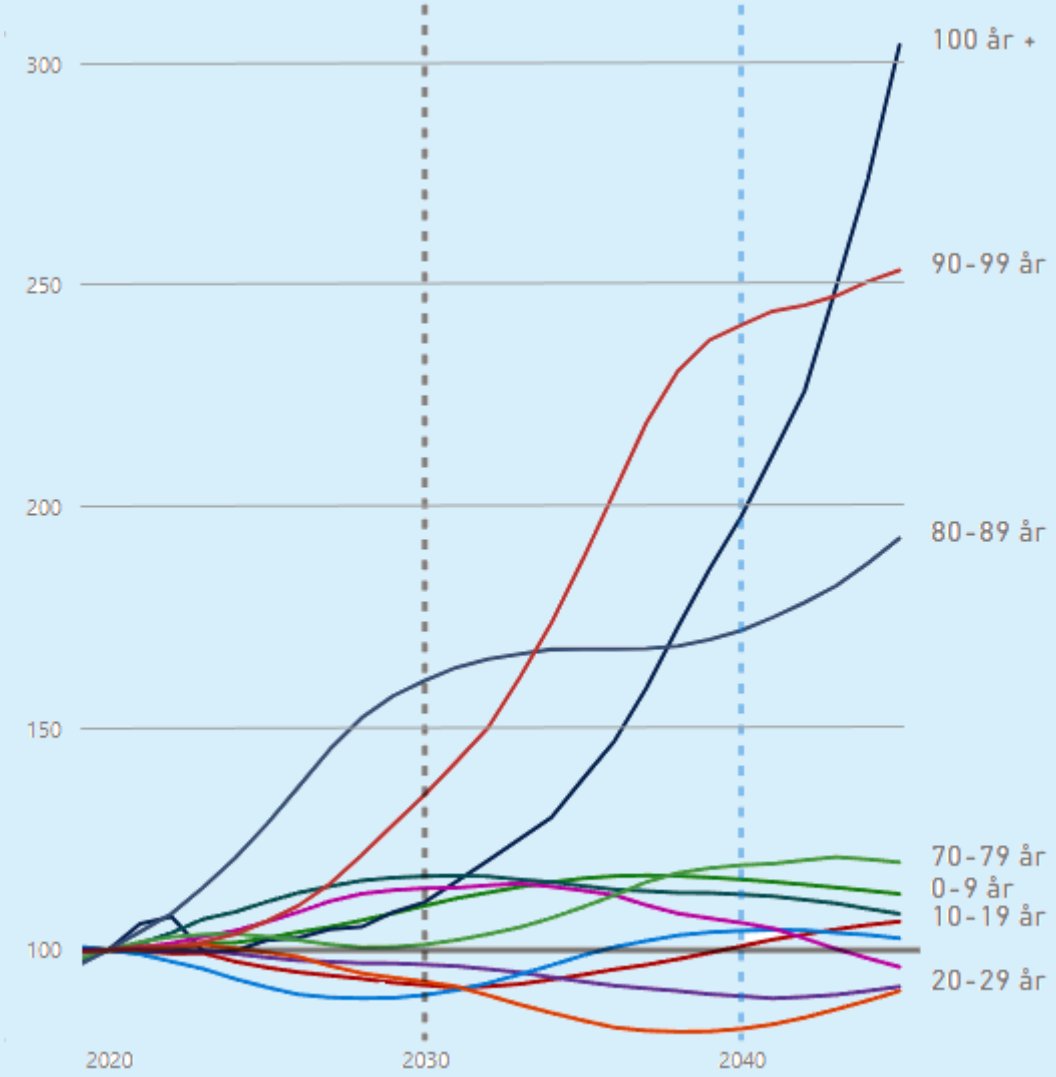
6. See improvement

- Maintain good initiatives
- Continuous focus

Data’s role in organization

- Data as **tool**
  - Data Driven Management
- Data as **commodity**
  - Data sharing = trust ==> collaboration
- Data as **practice**
  - Organizational learning = Optimize workflows
- Data as **algorithmic intelligence**
  - Process automation ==> SP all the time - example: list of most likely no-shows

Issue – demographic development



Source: The Future Rigshospital
Demographic projection based on numbers from Statistics Denmark

Denmark has a growing population of elderly and sick people while at the same time having a declining/stagnating population of working age.

It is a necessity to investigate how to:

1. Make innovative solutions (AI, digitalization)
2. Make the most out of the available resources
   - Capacity maximization
   - Minimize double work

All while benefitting the individual patient and the overall healthcare system

Operation rooms

All surgical procedures that are performed in an operating room (OR) – can both be day surgery and hospitalized patients.

Examples:

- Open heart surgery
- Gender affirming surgery
- Transplantations
- Cancer treatment
- Brain surgery
- Eye surgery

!!OBS!!
All data is in Danish

Questions

What are the main pin-points that we should work with to have a better use of our operating rooms (OR)?

Available data about….

- Time stamps for all activities in OR
- Procedures and diagnosis
- Personnel & resources
- Cancelled appointments

Anything showing a pattern or tendency will be of interest.

The goal is to create insights on visual displays that can be used and understood by clinical personnel on shift.

Our **challenge to you** is considering how complex data analysis can be shown, explained or visualized in an extremely easy-to-understand way.

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Method - examples

- **Key factor analysis** – identify top 5 factors affecting operations that goes over time or that describes operations that are cancelled
- **Clustering/K-means** – identify number K with greatest fit and show how operation activities are clustered on selected features
- **Principal component analysis (PCA) for dimensionality reduction** – what dimensions in the operating time schedule are essential for describing the OR activity and what are redundant?

We give you access to 2 years of OR data

~ 95.000 patients

~ 130.000 operations

~ 200.000 rows of data

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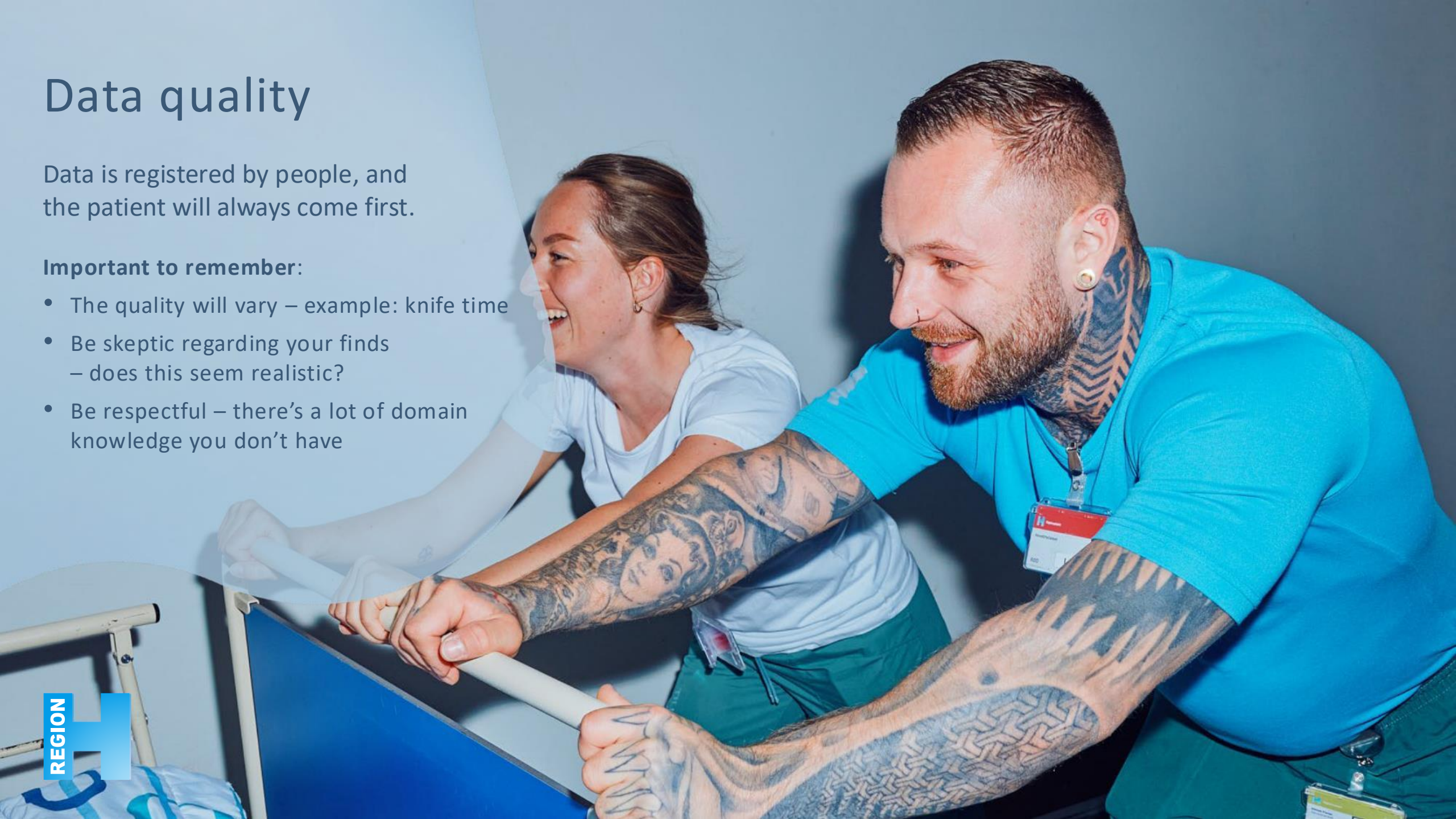
Data quality

Data is registered by people, and the patient will always come first.

Important to remember:

- The quality will vary – example: knife time
- Be skeptic regarding your finds – does this seem realistic?
- Be respectful – there's a lot of domain knowledge you don't have



Real people

Please show respect when you are handling data regarding real people in real life.

Requirements:

- Data is anonymized
- Analyze tendencies, not individuals
- Do not share the data with anyone outside the course
- Delete all data files after the course is completed

Thank you